

SECTION A

Answer **all** questions.

1. The first law of thermodynamics may be written as:

$$\Delta U = Q - W$$

(a) State the meaning of: [3]

ΔU

Q

W

(b) A closed cylinder fitted with a leak-proof piston as one end contains 0.060 mol of an ideal monatomic gas. The volume of the system is $1.4 \times 10^{-3} \text{ m}^3$ and the pressure of the gas is 100 kPa. Determine the temperature of the gas. [2]

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(c) The gas is heated and expands at constant pressure to a volume of $2.0 \times 10^{-3} \text{ m}^3$. Determine:

(i) the work done by the gas, [2]

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(ii) the increase in the internal energy of the gas, [2]

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(iii) the heat transferred to the gas. [1]

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